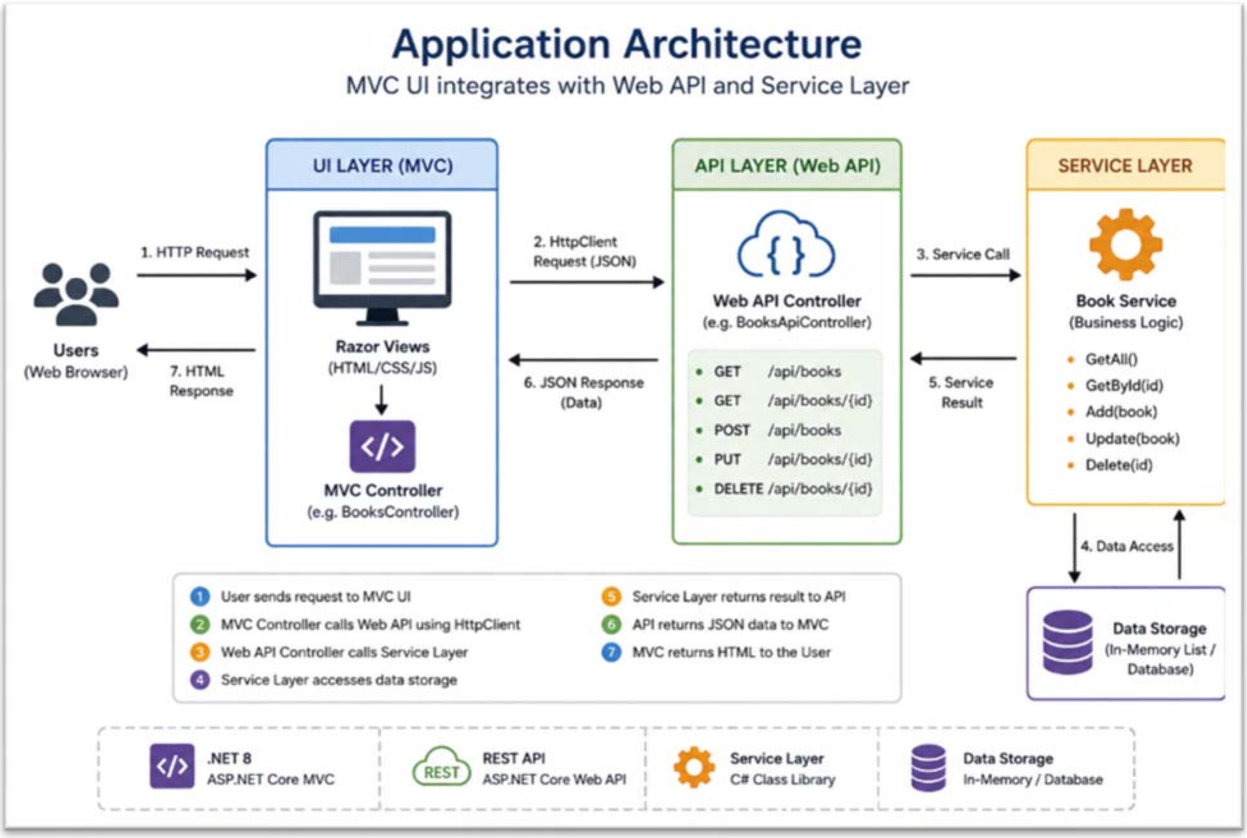
Lab: MVC + Web API Integration in ASP.NET Core

Project Description: Movie Review Portal

A university wants to create a platform where students can share books they want to exchange with other students. The application allows users to:

- View available books
- Add a new book
- View book details
- Remove a book from the list



1. Create a new MVC application:

```
dotnet new mvc -n BookExchangePortal
cd BookExchangePortal
dotnet run
```

2. Create the Book Model

```
namespace BookExchangePortal.Models
{
   public class Book
   {
      public int Id { get; set; }

      public string Title { get; set; } = "";

      public string Author { get; set; } = "";

      public string Category { get; set; } = "";

      public string Owner { get; set; } = "";
   }
}
```

3. Create the Service Layer: add Services folder and then a class BookService

```
using BookExchangePortal.Models;

namespace BookExchangePortal.Services
{
   public class BookService
   {
      private static List<Book> books = new()
      {
         new Book
         {
            Id = 1,
            Title = "Clean Code",
            Author = "Robert Martin",
            Category = "Programming",
            Owner = "Ahmed"
         },

         new Book
         {
            Id = 2,
            Title = "Design Patterns",
            Author = "GoF",
            Category = "Software Engineering",
```

```
            Owner = "Sara"
        }
    };

    public List<Book> GetAll()
    {
        return books;
    }

    public Book? GetById(int id)
    {
        return books.FirstOrDefault(b => b.Id == id);
    }

    public void Add(Book book)
    {
        book.Id = books.Max(b => b.Id) + 1;
        books.Add(book);
    }

    public void Delete(int id)
    {
        var book = GetById(id);

        if (book != null)
            books.Remove(book);
    }
  }
}
```

4. Register Services: Open Program.cs

```
builder.Services.AddHttpClient();
builder.Services.AddSingleton<BookService>();
```

5. Create Web API Controller: Controllers/Api/BooksApiController.cs

```
using Microsoft.AspNetCore.Mvc;
using BookExchangePortal.Models;
using BookExchangePortal.Services;

namespace BookExchangePortal.Controllers.Api
{
  [ApiController]
  [Route("api/books")]
  public class BooksApiController : ControllerBase
  {
    private readonly BookService _service;
```

```
        public BooksApiController(BookService service)
        {
            _service = service;
        }

        [HttpGet]
        public IActionResult GetBooks()
        {
            return Ok(_service.GetAll());
        }

        [HttpGet("{id}")]
        public IActionResult GetBook(int id)
        {
            var book = _service.GetById(id);

            if (book == null)
                return NotFound();

            return Ok(book);
        }

        [HttpPost]
        public IActionResult AddBook(Book book)
        {
            _service.Add(book);

            return CreatedAtAction(
                nameof(GetBook),
                new { id = book.Id },
                book);
        }

        [HttpDelete("{id}")]
        public IActionResult DeleteBook(int id)
        {
            _service.Delete(id);

            return NoContent();
        }
    }
}
```

6. Test the API. Run the application and test: https://localhost:xxxx/api/books
7. Create MVC Controller: Controllers/BooksController.cs

```
using Microsoft.AspNetCore.Mvc;
using BookExchangePortal.Models;
using System.Net.Http.Json;

namespace BookExchangePortal.Controllers
{
    public class BooksController : Controller
    {
        private readonly IHttpClientFactory _factory;

        public BooksController(IHttpClientFactory factory)
        {
            _factory = factory;
        }

        public async Task<IActionResult> Index()
        {
            var client = _factory.CreateClient();

            var books =
                await client.GetFromJsonAsync<List<Book>>
                ("http://localhost:5125/api/books");

            return View(books);
        }
    }
}
```

8. Create MVC View

```
@model List<BookExchangePortal.Models.Book>

<h2>Available Books</h2>

<table class="table table-bordered">
    <thead>
        <tr>
            <th>Title</th>
            <th>Author</th>
            <th>Category</th>
            <th>Owner</th>
        </tr>
    </thead>

    <tbody>
        @foreach(var book in Model)
        {
```

```
        <tr>
          <td>@book.Title</td>
          <td>@book.Author</td>
          <td>@book.Category</td>
          <td>@book.Owner</td>
        </tr>
      }
    </tbody>
</table>
```

9. Add Navigation Menu: open Views/Shared/_Layout.cshtml

```
<li class="nav-item">
    <a class="nav-link"
      asp-controller="Books"
      asp-action="Index">
      Books
    </a>
</li>
```

Questions:

- Create a Book Details page
- Create a New Book Form
- Submit the form to the Web API using: POST /api/books
- Add validation: Title and Auhor are required
- Create a Delete button that calls: DELETE /api/books/{id}